Capstone Project Proposal: Synthetic Data Generation for Construction ML Models

1. Project Overview

Suffolk's Advanced Analytics & Data Science team is increasingly adopting state-of-the-art machine learning techniques to power predictive, planning, and generative capabilities across the construction lifecycle. However, these models depend on large, high-quality datasets—something the current Platinum analytics layer cannot yet provide at meaningful scale.

This capstone project will design and implement a **Synthetic Data Generation Framework** capable of producing large volumes of statistically valid construction project data across Suffolk's curated Platinum schema. The resulting synthetic dataset will unlock experimentation with advanced ML architectures, accelerate model development, and help validate future DaaS product concepts—without relying solely on limited historical data.

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2. Current Problem

Limited Data Availability

- The Platinum layer contains ~200 fully modeled, end-to-end projects.
- Many fields are partially complete, inconsistently logged, or contain long gaps.
- Small sample sizes create difficulty training ML models that rely on statistical stability.
- Leading ML approaches (e.g., deep learning, transformers, diffusion models) typically require **large, rich datasets** to achieve reliable performance.
- Teams routinely hit the ceiling of available real data, slowing innovation and experimentation.

Opportunity: Synthetic Data

- Generate **thousands of statistically realistic synthetic projects** aligned with true distributions, correlations, and business rules found in Platinum.
- Use generative modeling (e.g., CTGAN, Gaussian copulas, diffusion models) to produce structured, tabular project and schedule-related fields.
- Leverage **LLMs to generate associated unstructured data** (e.g., Observations, RFIs, risk narratives, daily logs) that reflect real construction context.
- Expand the effective dataset size by >10×, enabling advanced ML experimentation while avoiding privacy, confidentiality, or availability constraints.

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3. Project Scope

3.1 Data Domains Included

Synthetic generation will focus on the *top value curated models* within Suffolk's Platinum schema—likely including:

- **Project-level attributes** (size, geography, sector, delivery method)
- **Schedule summary metrics** (duration, critical path indicators, % complete patterns)
- **Change events / risk markers**
- **Safety indicators** (lagging and leading metrics)
- **Cost-related aggregates** (high-level, non-sensitive fields)
- **Resource and staffing patterns**

Where possible, the generator will respect:

- Real-world correlations (e.g., project size ↔ budget ↔ duration)
- Valid ranges and categorical distributions
- Time-series or sequential characteristics for schedule curves

3.2 Text / Descriptive Data

Using fine-tuned or prompt-steered LLMs, generate synthetic versions of:

- **Observations** (site conditions, quality notes, daily risks)
- **RFIs and issue summaries**
- **Narrative insights** used in tripwire explanations, planning documentation, or weekly reporting

Models will be designed to mimic tone, structure, and semantic content—while ensuring no memorization of actual Suffolk text.

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4. Proposed Methodology

Phase 1: Data Audit & Schema Prioritization

- Evaluate completeness and distributions of Platinum tables.
- Select target entities for synthetic replication.
- Identify required business-rule constraints.

Phase 2: Model Selection & Prototype

- Evaluate multiple generative approaches:
  - Variational Autoencoders (VAEs)
  - CTGAN / TVAE (for tabular synthesis)
  - Gaussian Copula models
  - Diffusion-based generators for structured data
- Benchmark models against fidelity, privacy, correlation preservation, and scalability.

Phase 3: Text Synthesis (LLM-Powered)

- Design prompts or fine-tuning sets to generate realistic construction narratives.
- Ensure semantic coherence without leakage of real sensitive text.

Phase 4: Quality Evaluation Framework

- Distribution similarity metrics (KS test, χ^2 tests)
- Correlation preservation
- Train-on-synthetic / test-on-real performance lift
- Privacy metrics (distance to nearest records, membership inference)

Phase 5: Deliverables & Deployment

- A reproducible synthetic data pipeline (Python/R) integrated with Databricks.
- A cataloged synthetic dataset with 1,000–10,000+ projects.
- Documentation and dashboards describing evaluation metrics.
- Recommendations for downstream ML experimentation (e.g., schedule models, safety prediction, risk classification).

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5. Expected Impact

Accelerates ML Innovation

- Removes the data-scarcity bottleneck for generative schedule modeling, safety forecasting, and tripwire development.

Supports Product-Led DaaS Strategy

- Helps validate future analytics and ML products before large real-world datasets are available.

- Enables rapid prototyping and “what-if” experimentation for emerging models.

Reduces Dependency on Real Project Data

- Synthetic data avoids confidentiality issues and provides unlimited scalability.
- Provides safer datasets for vendor collaboration, sandboxing, and internal training.

Strengthens Model Robustness

- Larger, more diverse training data improves generalization and reduces overfitting in ML pipelines.

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6. Capstone Deliverables

- **Synthetic Data Generator** (code + configuration)
- **Curated Synthetic Dataset** aligned to Platinum schema
- **Evaluation & Validation Report** (accuracy, fidelity, privacy)
- **LLM-Generated Text Library** for Observations, RFIs, narratives, etc.
- **Demo Notebook or Shiny Dashboard** showcasing dataset characteristics
- **Executive Summary** describing business value and recommended next steps

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7. Conclusion

This capstone project will provide Suffolk with a foundational capability for **scalable, private, high-fidelity synthetic construction data**—dramatically expanding what the Data Science team can build, experiment with, and validate. The deliverables will directly support ongoing ML initiatives, as well as enabling future ones.